

Software Engineering and Data Science
SEIS 766: Vision AI
Assignment #8

Due Date: April 9th

1. Read 300 numeric vectors from “VAI_HW_CLIP_Image_Vecs.csv”. Each vector has 1568 dimensions and belong to a true class Y . The first 100 vectors belong to class 0, the second 100 vectors belong to class 1, and the last 100 vectors belong to class 2. Use all the 300 vectors for training and testing in this assignment (i.e. **NO** need to split into training / testing sets). Also, **NO** need to standardize vectors as all the predictors have the same scale.
2. Train a classifier C (with dense layers) to classify those 300 vectors into their classes. (HINT: do not overfit or underfit your classifier C)
3. Input the 300 vectors to the classifier C to obtain their predicted classes $Yhat1$.
4. Build a classification report based on the 300 true class Ys and $Yhat1$.
5. Train a VAE using all the 300 vectors.
6. Randomly pick 30 vectors and their corresponding true classes Y from the input 300 vectors as testing vectors. Input the 30 testing vectors into VAE to obtain 30 “reconstructed” vectors.
7. Input the 30 “reconstructed” vectors to the classifier C to obtain their predicted classes $Yhat2$.
8. Build a classification report based on the 30 true class Ys and $Yhat2$.
9. Include the two classification reports in your WORD document.

Submission Guideline:

1. Please include the answers to the above questions in a WORD document.
2. Please print your programs (matlab or python) as **PDF / html** and include the **PDF / html** in your submission.
3. Please include *****ALL***** your programs a8.m/.mlx/.py/.inpyb in your submission.
4. Prepare EVERYTHING mentioned in the guideline and submit them on **Canvas** no later than the due date. Please do **NOT** zip your files.
5. Please carefully follow the submission guideline. Otherwise, the instructor may not be able to grade your assignment.